

Transition from IPv4 to IPv6

Computer Network Technologies and Services



Copyright Notice

- This set of transparencies, hereinafter referred to as slides, is protected by copyright laws and provisions of International Treaties. The title and copyright regarding the slides (including, but not limited to, each and every image, photography, animation, video, audio, music and text) are property of the authors specified on page 1.
- The slides may be reproduced and used freely by research institutes, schools and Universities for non-profit, institutional purposes. In such cases, no authorization is requested.
- Any total or partial use or reproduction (including, but not limited to, reproduction on magnetic media, computer networks, and printed reproduction) is forbidden, unless explicitly authorized by the authors by means of written license.
- Information included in these slides is deemed as accurate at the date of publication. Such information is supplied for merely educational purposes and may not be used in designing systems, products, networks, etc. In any case, these slides are subject to changes without any previous notice. The authors do not assume any responsibility for the contents of these slides (including, but not limited to, accuracy, completeness, enforceability, updated-ness of information hereinafter provided).
- In any case, accordance with information hereinafter included must not be declared.
- In any case, this copyright notice must never be removed and must be reported even in partial uses.



The road to IPv6



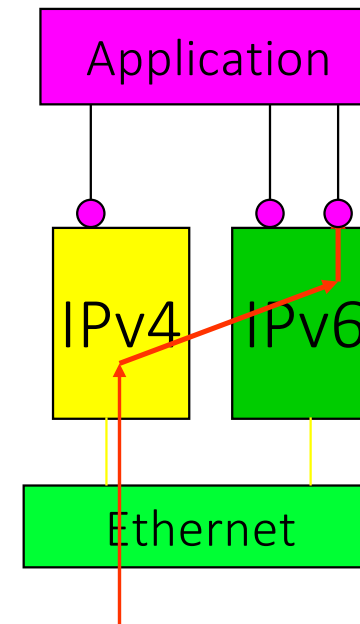
IPv4 to IPv6 transition

- Ideally:
 - Incremental
 - Seamless
 - Smooth

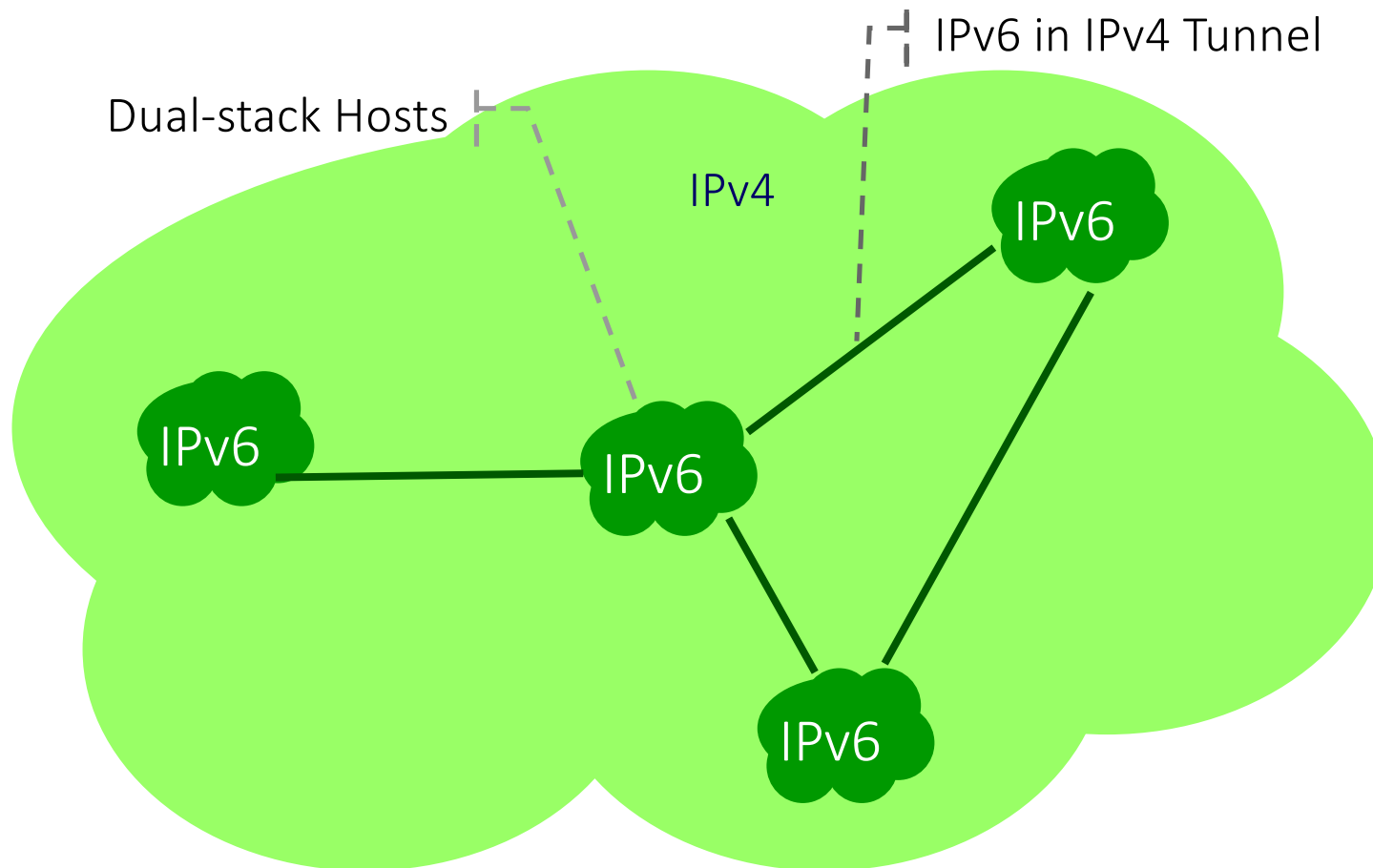


How can it be achieved?

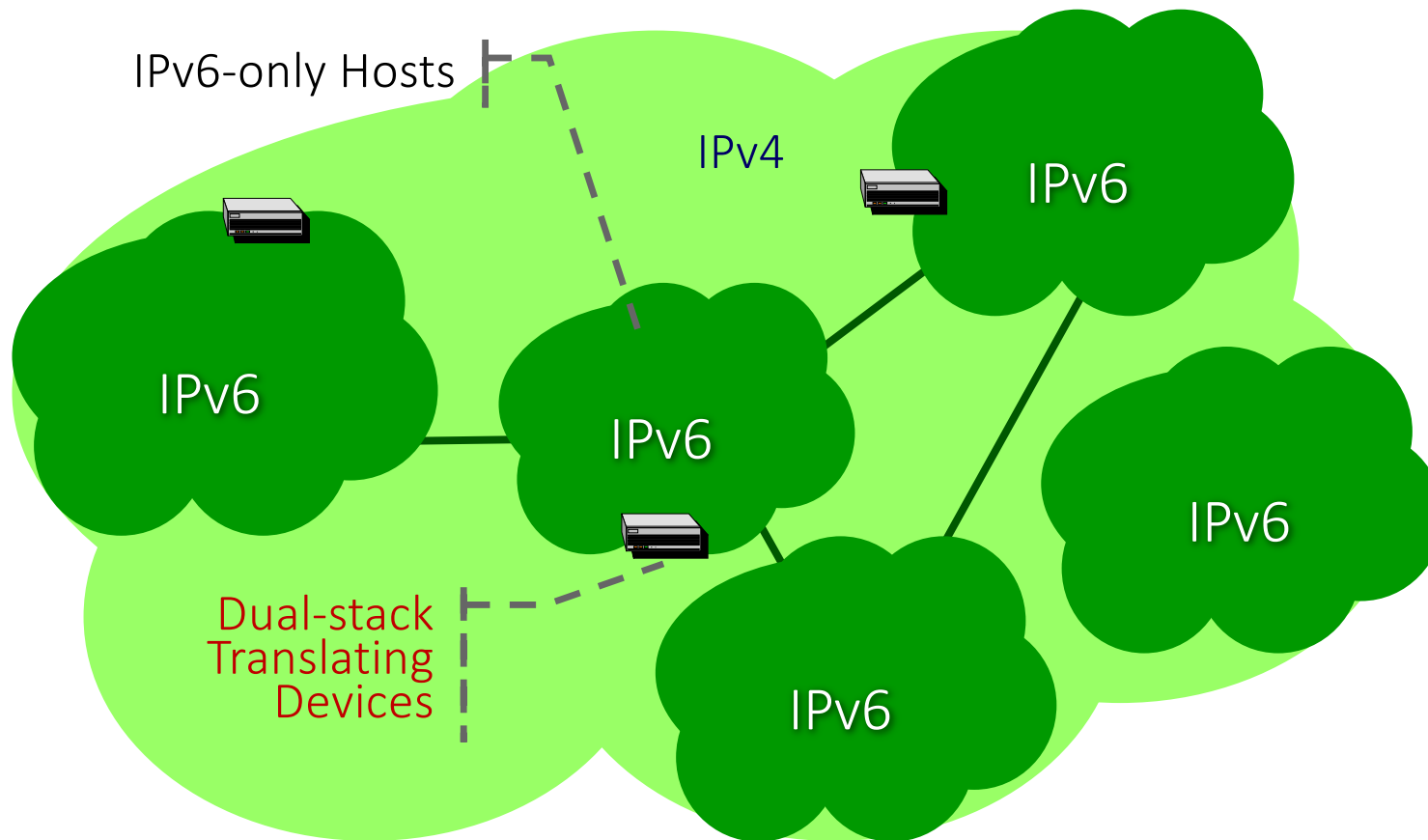
- Dual-stack approach
 - IPv6 as a new layer-3 protocol
 - Generate/receive v6 or v4 packets as needed
- Address mapping
- Tunneling
- Translation mechanisms



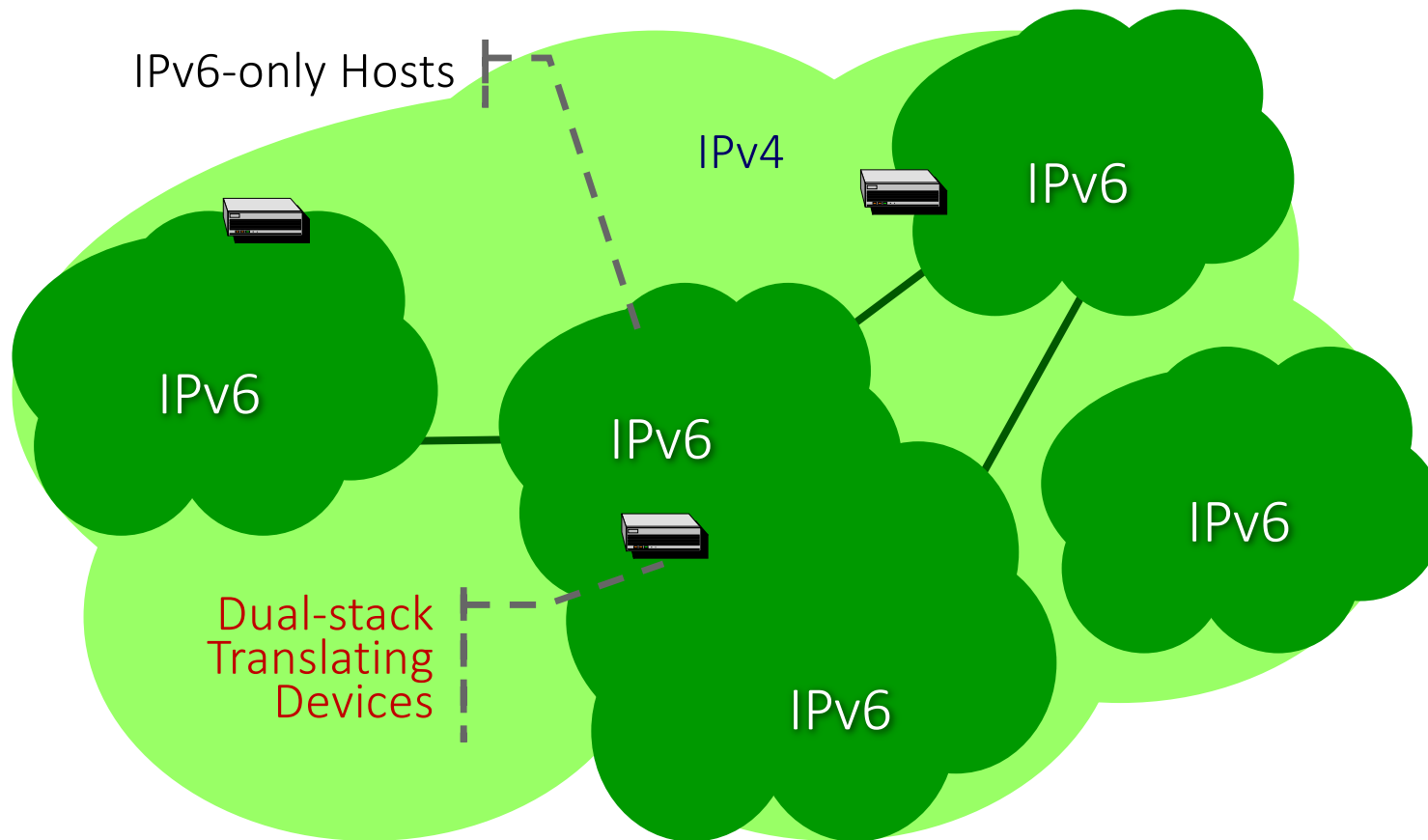
Step 1: isolated IPv6 networks



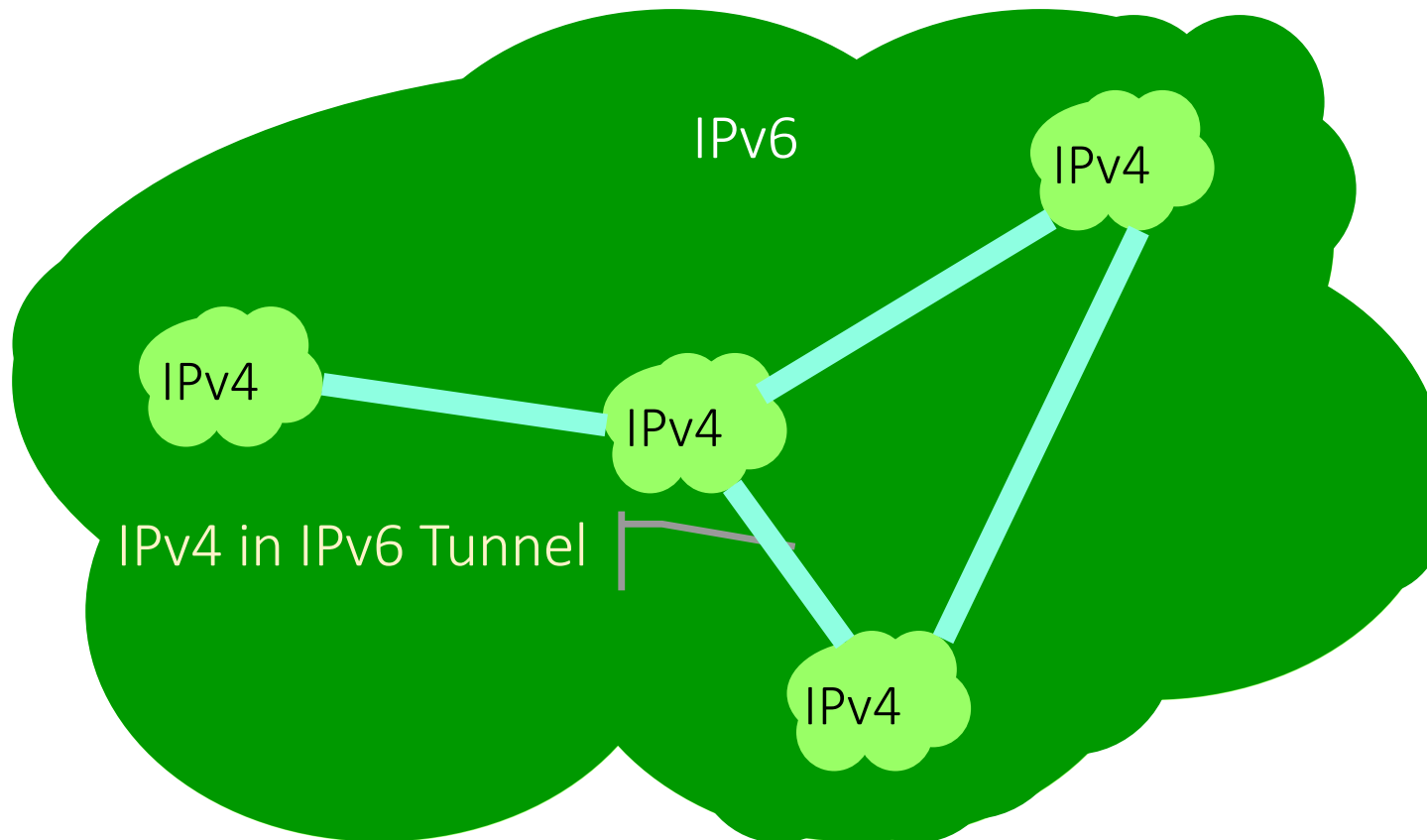
Step 2: IPv6 islands grow



Step 3: Native IPv6 connectivity



Step 4: IPv6 takes over



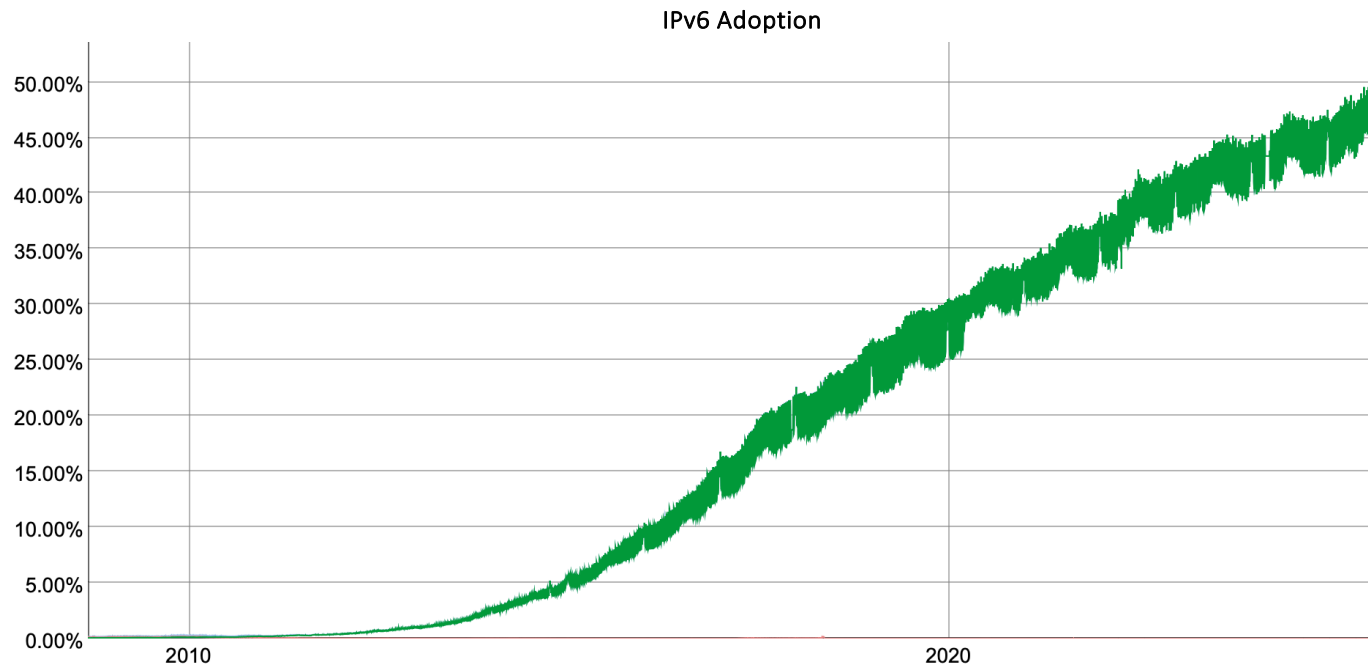
Are we ready?

- Quite so!
 - All protocols specified since 1996
- IPv6 implemented on **routers**
 - Even if less stable than IPv4
 - Possibly missing some functionalities
 - Also hardware implementations (Layer-3 switches)
- IPv6 implemented in **end systems**
 - Windows (since 2000 version)
 - Unix: Ubuntu, Debian, FreeBSD...
 - MAC (since MAC OS X)



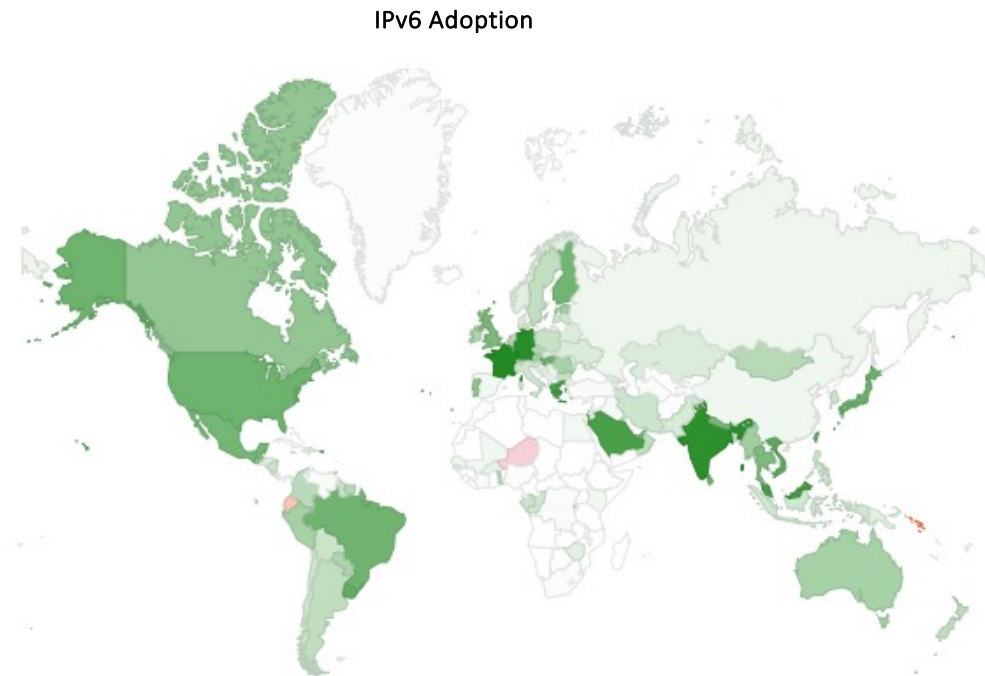
When will it happen?

- Large IPv4 install base
- As of October 2025, according to Google, the IPv6 adoption rate globally is around 50% (percentage of users that access Google over IPv6)



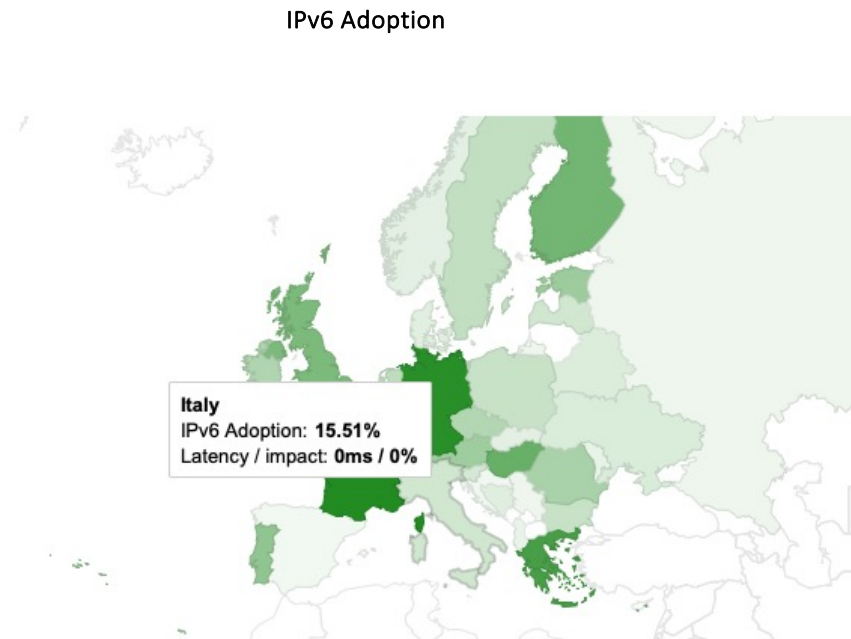
When will it happen?

- Large IPv4 install base
- As of October 2025, according to Google, the IPv6 adoption rate globally is around 50%
 - 54% in the U.S. alone!



When will it happen?

- Large IPv4 install base
- As of October 2025, according to Google, the IPv6 adoption rate globally is around 50%
 - 54% in the U.S. alone
 - 17% in Italy, >80% in France/Germany
- Only one true motivation:
Address space depletion
- no official switch-off date
 - Will happen gradually



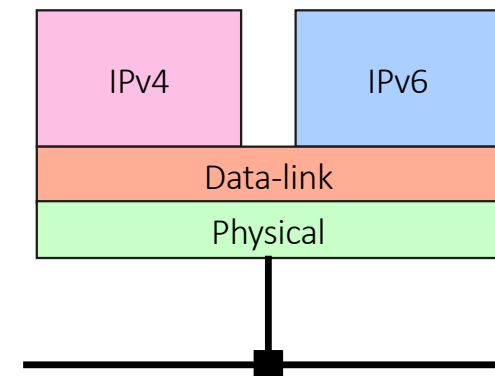
Assumption (and problem to solve)

IPv4 and IPv6 will coexist
(at least for a while)



Dual Stack Approach

- Both IPv4 and IPv6 capabilities
 - All hosts and routers supporting IPv6
 - IPv4 support can be (gradually) removed (and included in new hosts) once all hosts have IPv6
- Hosts communicate natively with both
- Complete duplication of all protocol stack components
 - Routing protocols
 - Routing table
 - Access lists



Dual Stack Limitations

- It does not reduce the need for IPv4 addresses
 - Each host still needs an IPv4 address to use IPv4
- Applications have the responsibility whether to use IPv4 or IPv6



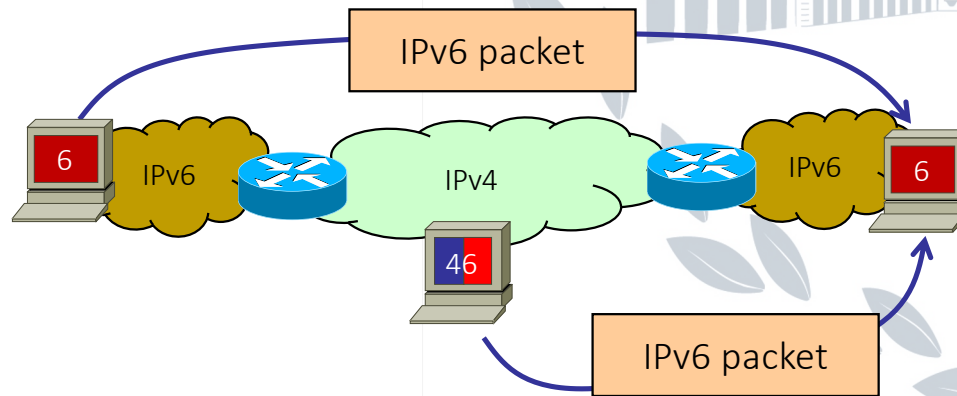
Not all hosts will be dual stack

- IPv6 hosts shall communicate with IPv6 hosts through an IPv4 network
 - Same for IPv4 hosts through an IPv6 network
- IPv6 hosts shall communicate with IPv4 hosts
 - Translation mechanisms must be used
 - Not targeting IPv4 hosts contacting IPv6 ones
 - Difficult to map the large IPv6 address space on smaller IPv4 address space



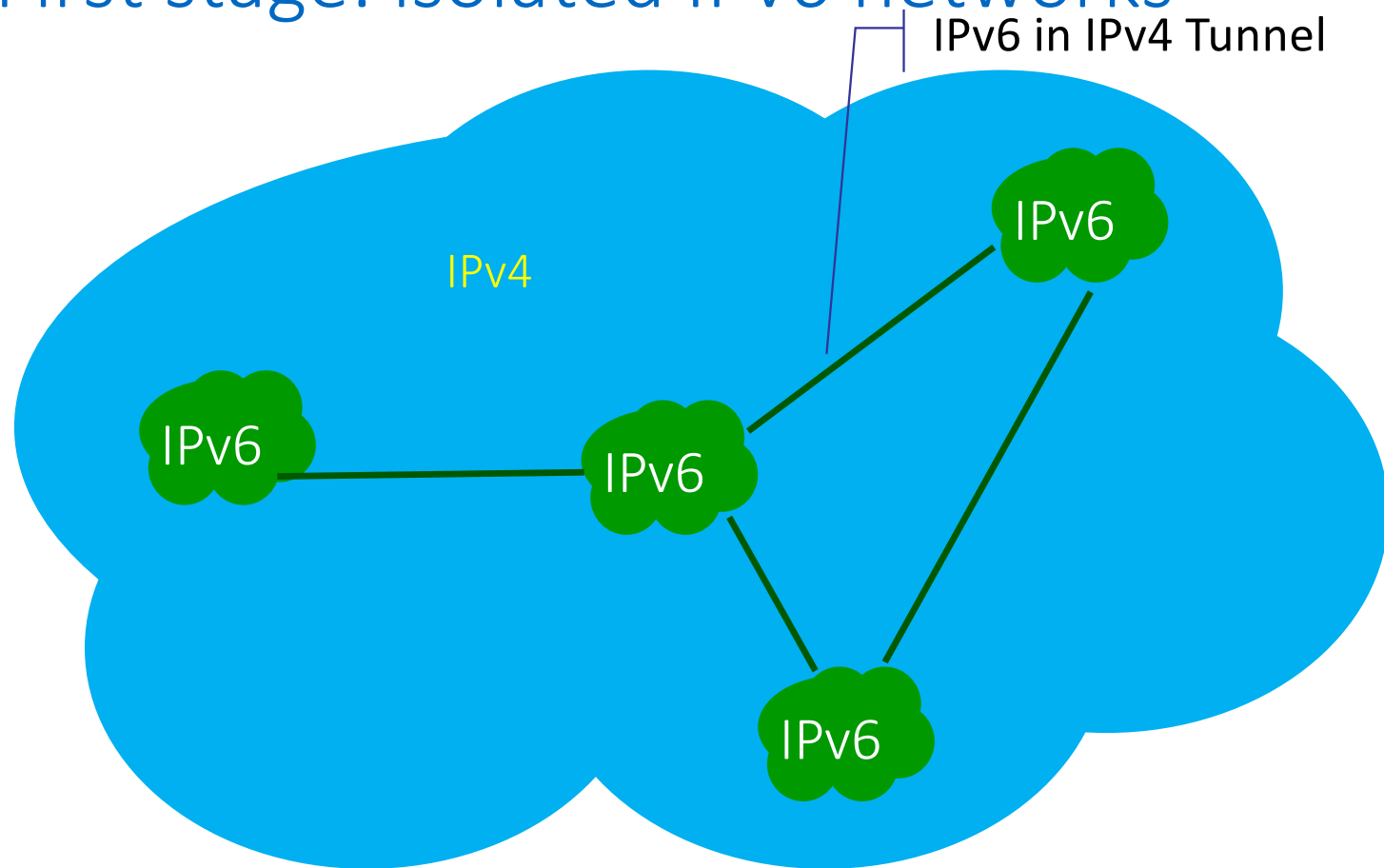
Tunneling Solutions

Dual stack and native IPv6 hosts exchange IPv6 packets through an IPv4 network



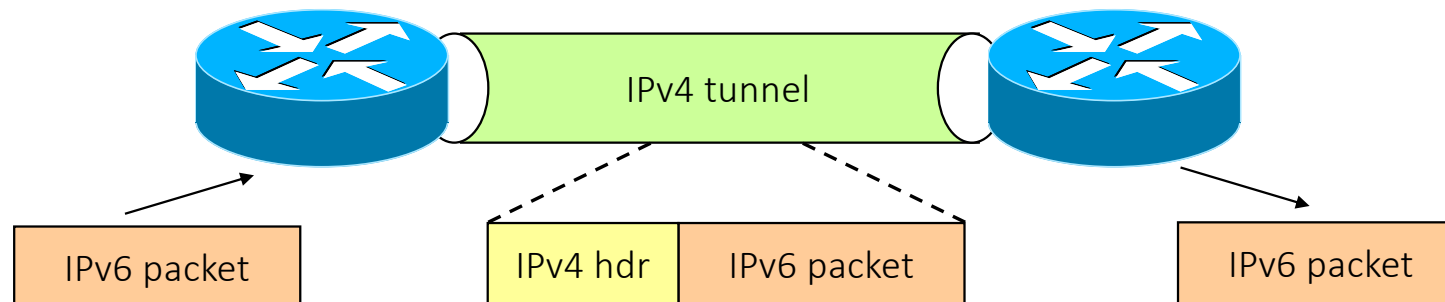
1859

First stage: isolated IPv6 networks



Traversing a IPv4-only Network

- Tunnelling
 - Encapsulation of IP (v6) packets into IP (v4) packets
- Emulates a “direct” link among IPv6 devices

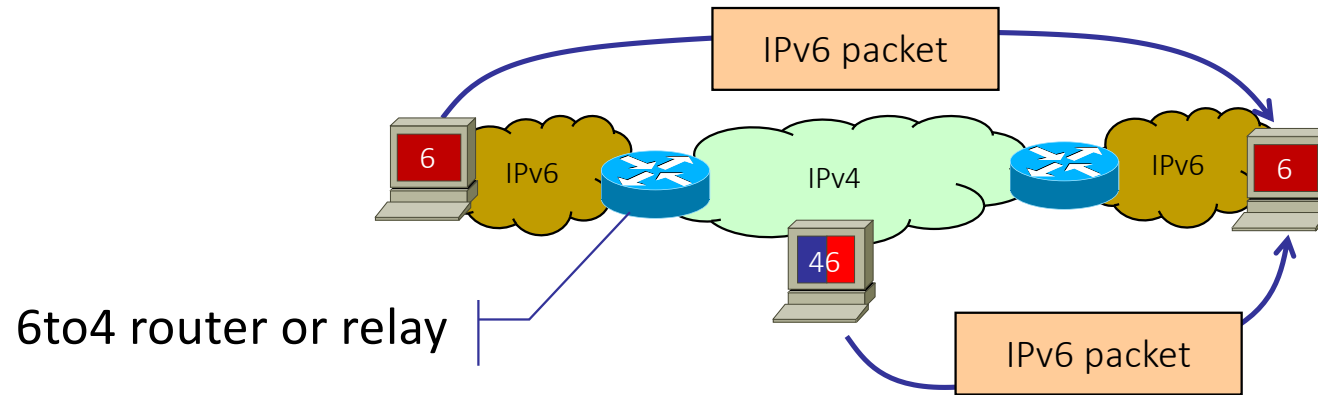


Tunnelling

- End points: hosts and routers
- Protocols
 - GRE (Generic Routing Encapsulation)
 - IPv6 in IPv4
 - Protocol type = 41
- Set up: manual and automatic
 - IPv4-compatible addresses, 6over4 (RFC 2529), 6to4, Tunnel Broker (RFC 3053), ISATAP, Teredo



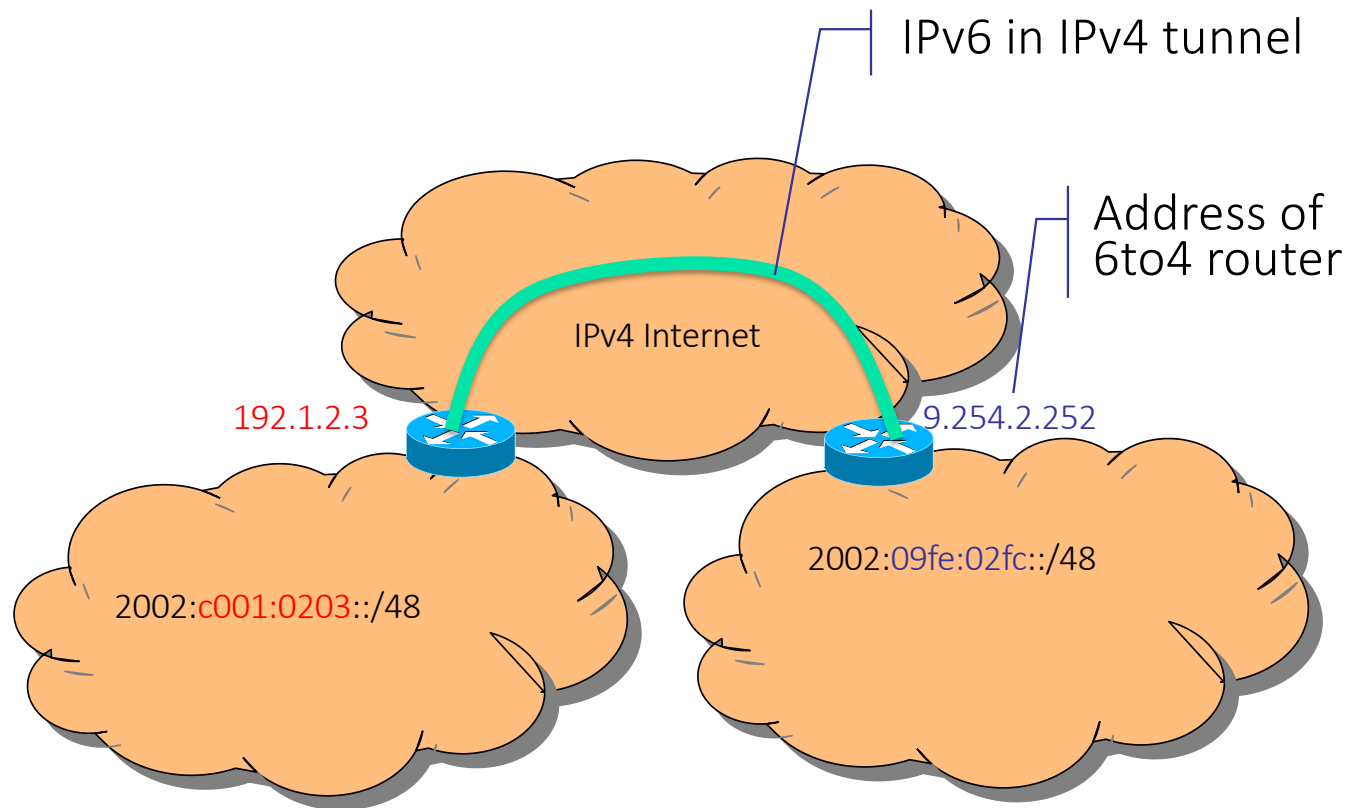
6to4



- Each 6to4 router is assigned a unique IPv6 address that begins with the prefix "2002::/16" followed by its 32-bit IPv4 address
- Packets sent by an IPv6 host are encapsulated in IPv4 by a 6to4 router and routed through IPv4 Internet to another 6to4 router
- The 6to4 router decapsulates it to reveal the original IPv6 packet and routes it to the destination

Basic 6to4 Scenario

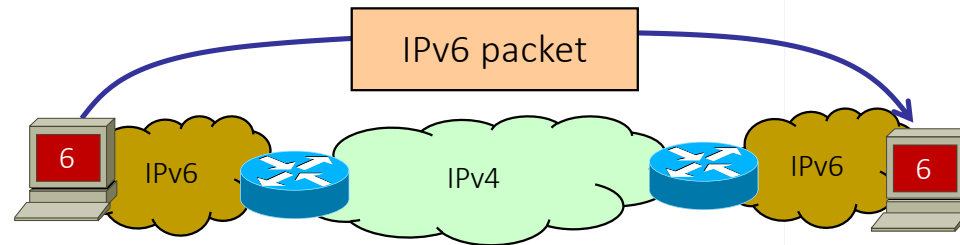
Not meant for communication between an IPv4 host and an IPv6 host



6to4 problems

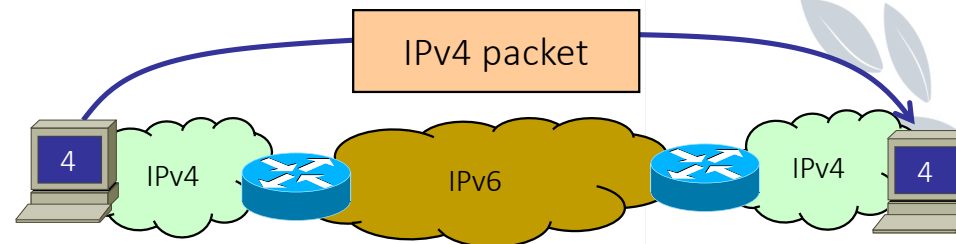
- 6to4 was progressively phased out starting from 2010 because it introduced complexity into the network configuration
 - address conflicts and misconfigurations (large IPv6 address pool mapped onto a smaller IPv4 pool)
 - not all networks had globally routable IPv4 addresses available
 - NAT and firewalls interfered with 6to4 deployment
 - security vulnerabilities, particularly related to routing and address spoofing





Scalable, Carrier-grade Solutions

Native IPv6 (IPv4) hosts exchange IPv6 (IPv4) packets through an IPv4 (IPv6) network



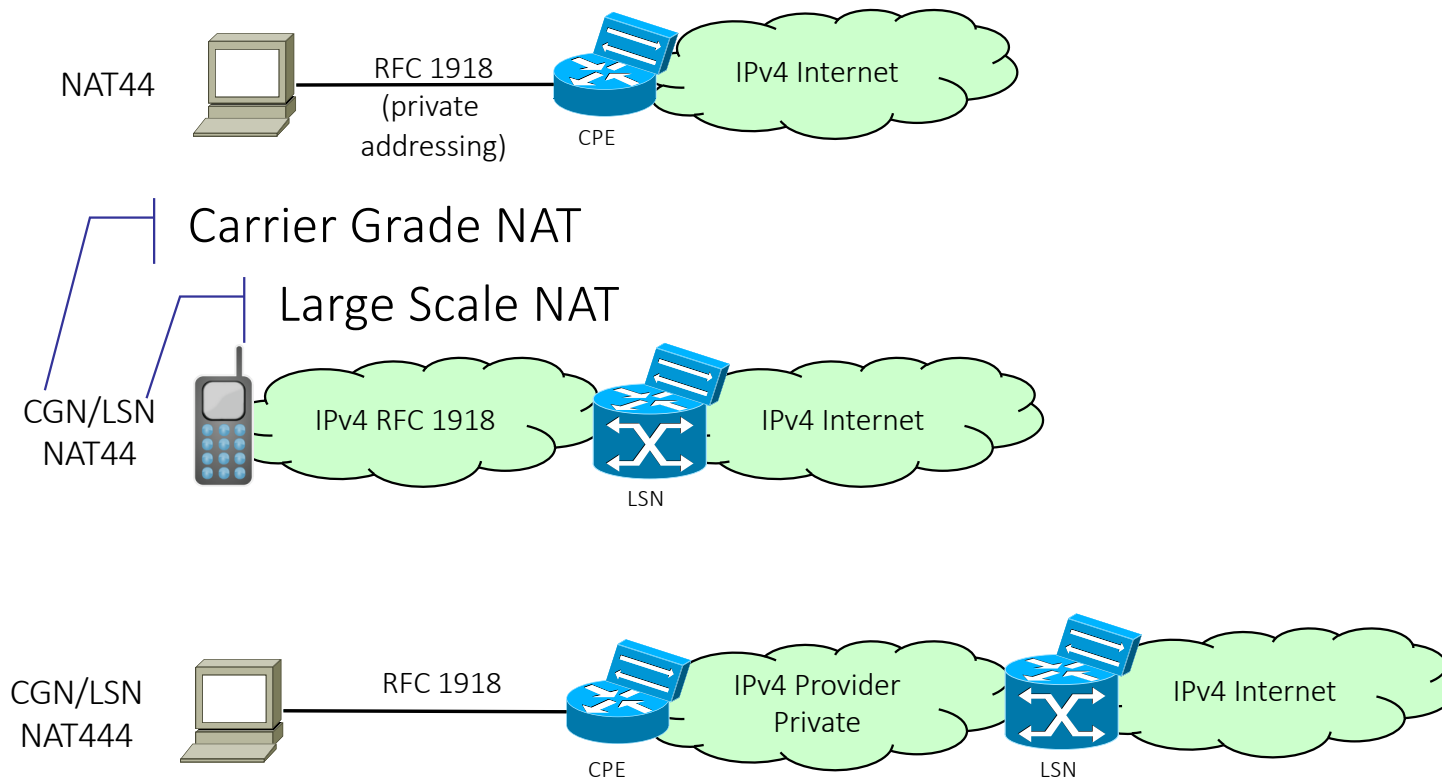
1859

Goals

- Still need to support
 - IPv4 servers possibly communicating with IPv6 hosts
 - IPv4 clients
- Several Options
 - DS-Lite
 - A+P (DS-Lite evolution)
 - NAT64
 - MAP-T and MAP-E
 - 6PE (MPLS-based)



NAT is widely used today



How do we like NAT?

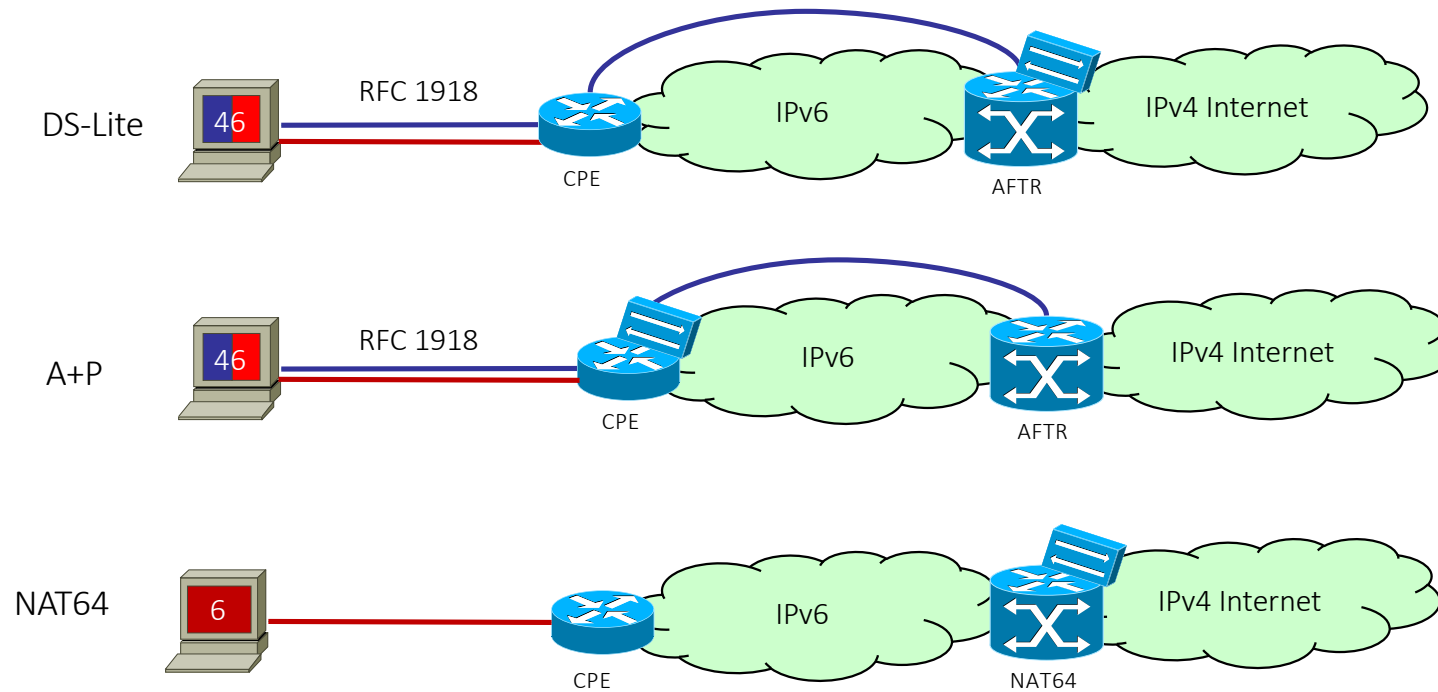
- Problematic with inbound sessions
 - E.g., servers
 - NAT + STUN/TURN may be ok for peer-to-peer sessions
- Bottleneck and single point of failure

Nevertheless

- Several (independent) cascaded instances of NAT are now very common
 - Starting from virtual machines
- Difficult to do without due to scarce addresses



Same Architecture with IPv6



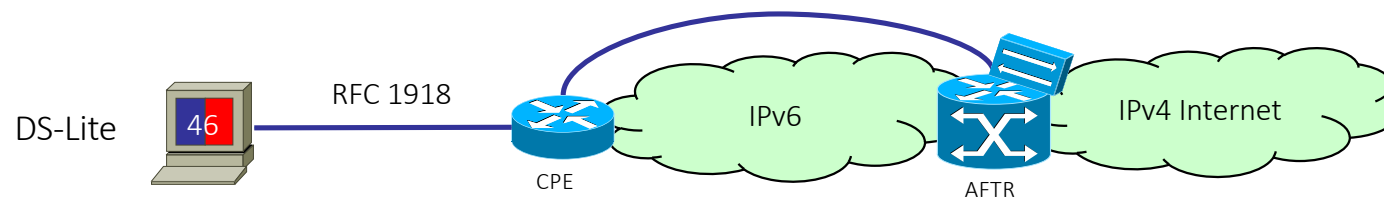
CPE: Customer Premises Equipment
AFTR: Address Family Transition Router

Note where the NAT function is located!



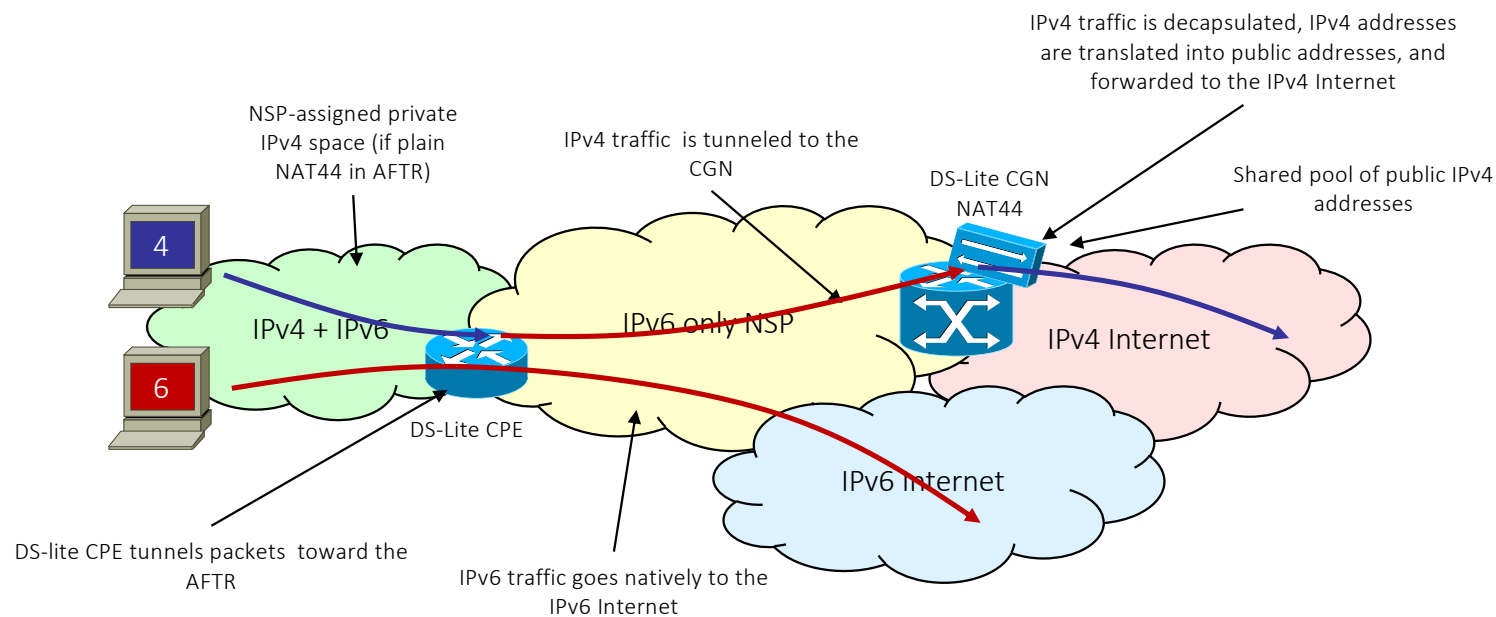
AFTR: Address Family Transition Router

- Allows IPv4 host to communicate with IPv4 hosts over an IPv6 carrier infrastructure
 - Residential host and current providers
- Features an IPv6 tunnel concentrator and possibly a large-scale NAT
- In use in DS-Lite and A+P



DS-Lite (Dual-Stack Lite)

- Dual-stack at the edge
- IPv6-only Service Provider backbone



Properties

- Reduces requirement for IPv4 addresses compared to dual-stack approach
 - Dual-stack requires public IPv4 address per host
- Extended NAT enables customer assigned (i.e., overlapping) addressing
 - IPv6 address of CPE in NAT table



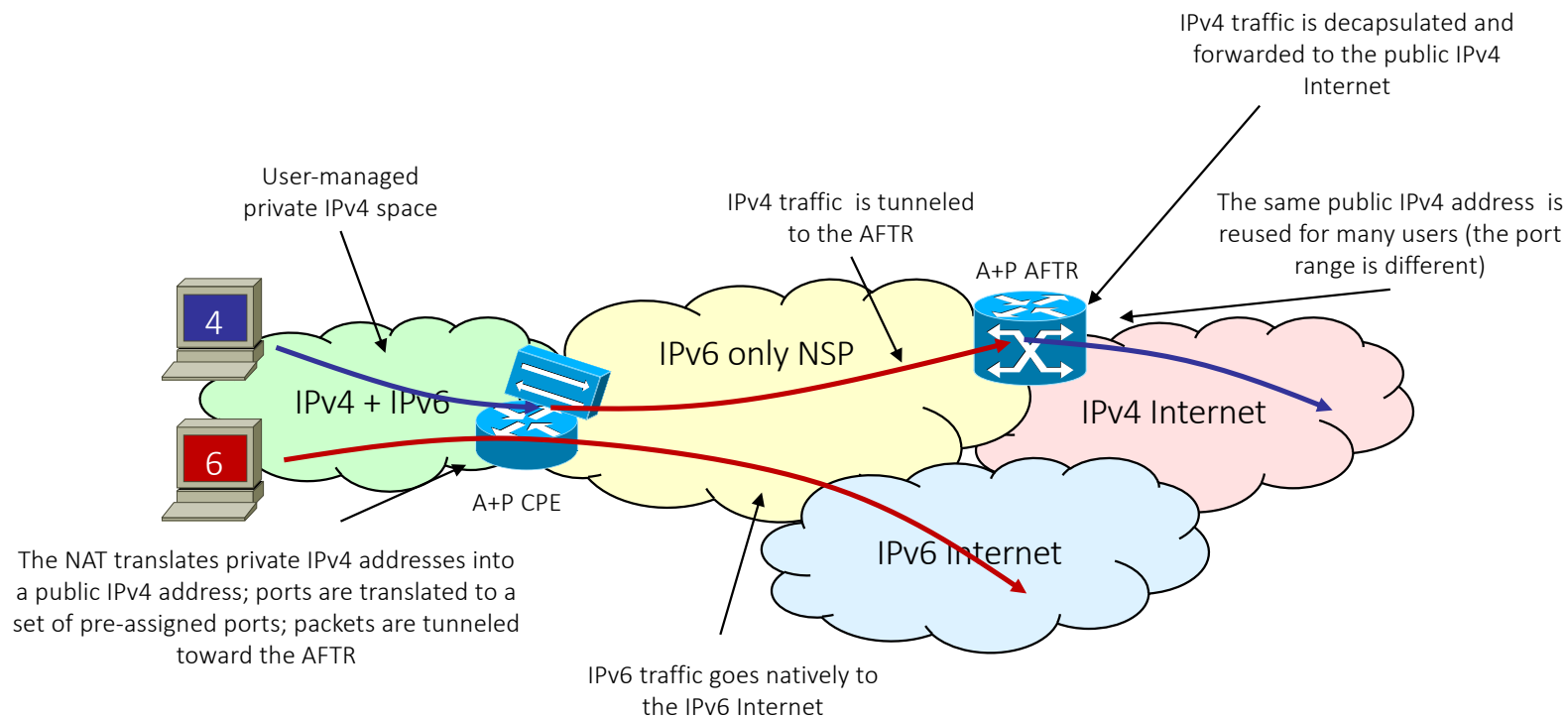
Limitations

- NAT is not under control of customer
 - Same problem as with CGN
- Problematic with servers
 - Static mapping and port forwarding cannot be configured



A+P (Address plus Port)

- NAT is under control of customer
- Ranges of TCP/UDP ports are assigned to each customer
 - Only ports used by NAT on outside

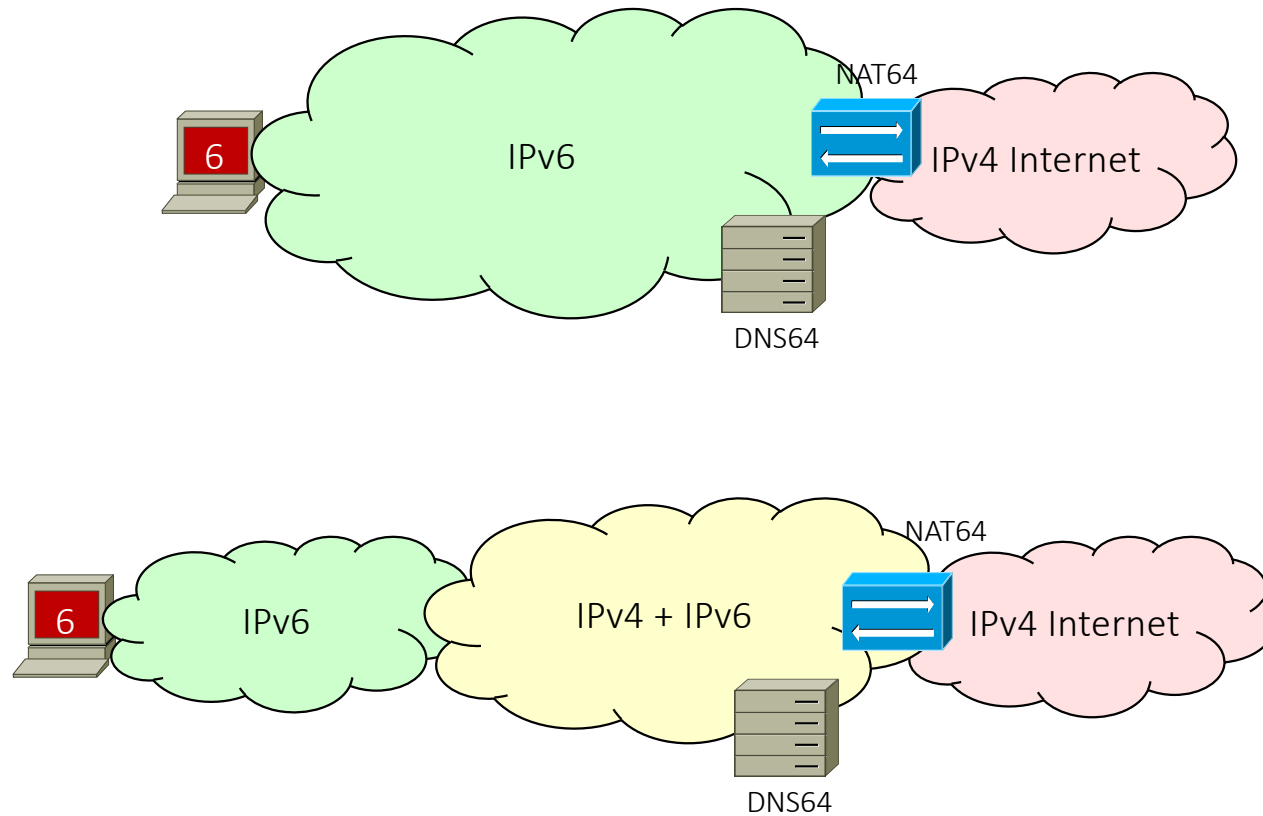


Features

- No problem with overlapping private address spaces at customers'
- Ports can be assigned automatically to CPE using the Port Control Protocol (PCP)
 - CPE can negotiate more ports any time
- AFTR is just a IPv4-in-IPv6 (proto-41) tunnel terminator
 - NAT44 is no longer needed in the AFTR



NAT64 + DNS64: Deployment Scenarios

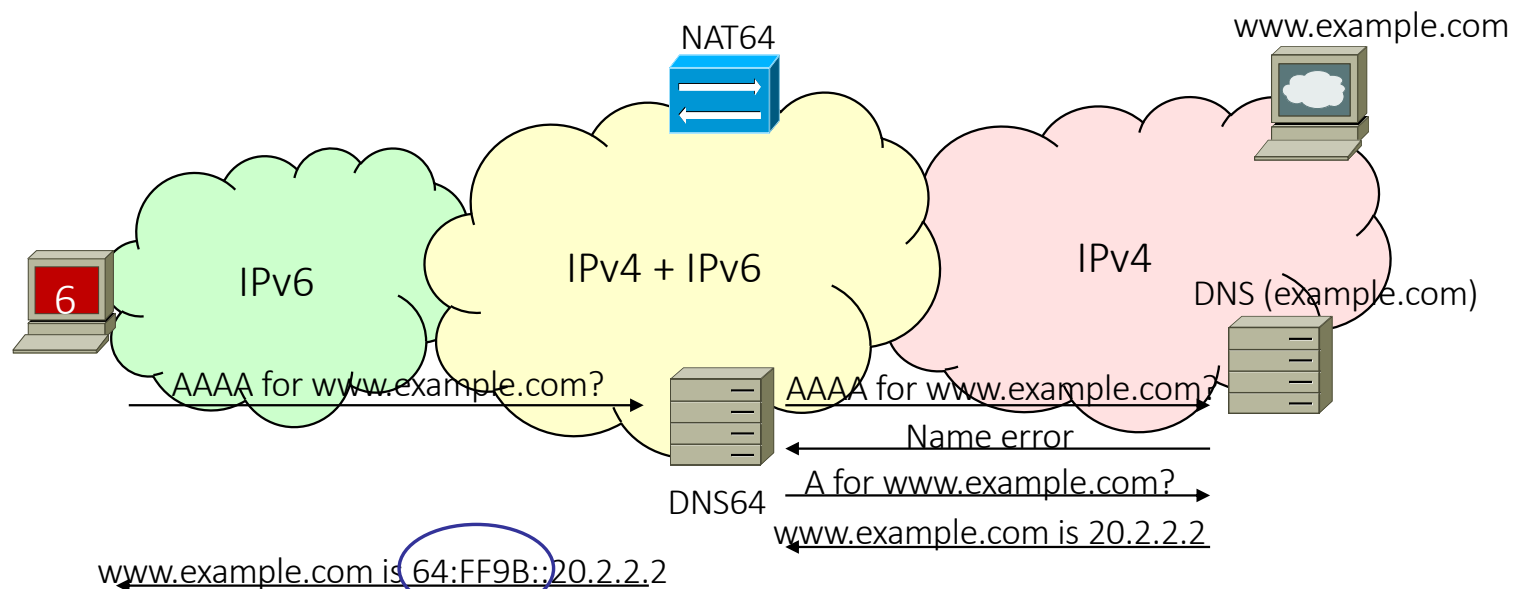


NAT64 + DNS64: Principles

- An IPv6 prefix is dedicated to mapped IPv4 addresses
 - Either well-known or network-specific
 - RFC 6052
- DNS64 maps A records into AAAA using NAT64 prefix, then serves A and AAAA records to the client
- NAT64 router advertises NAT64 prefix into IPv6 network to attract traffic toward IPv4 hosts



DNS64: Name Resolution

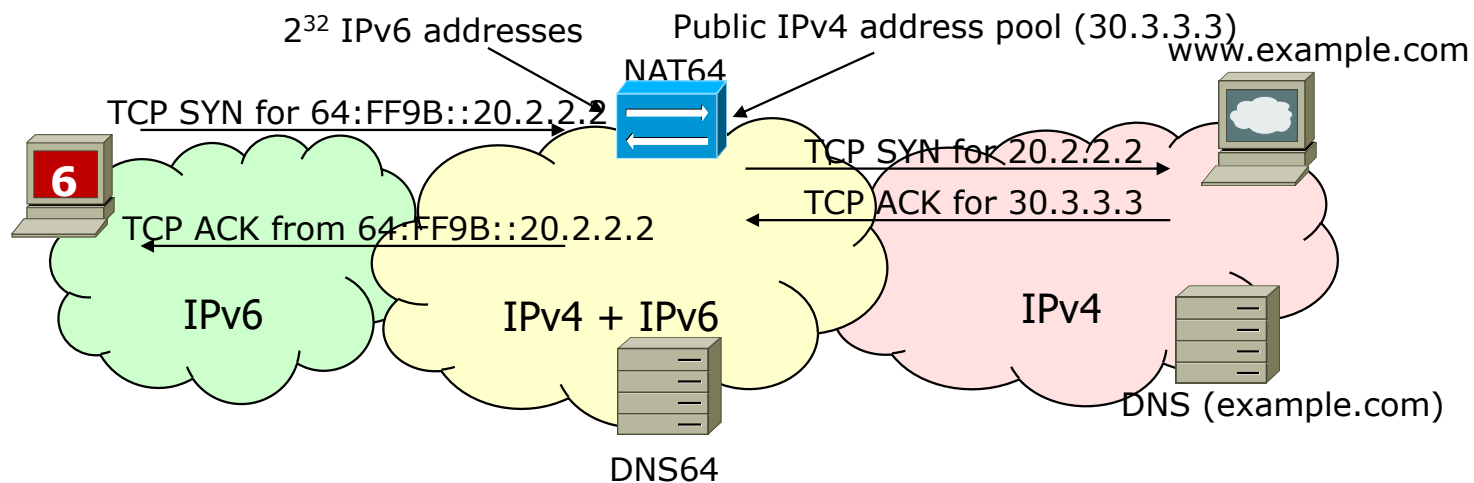


Well-known or deployment specific prefix (configured on DNS64 and possibly NAT64)



NAT64: Packet Forwarding

- NAT64 (outbound)
 - Translates IPv6 address and packet into IPv4
 - Picks a free IPv4 address/port from its pool
 - Builds NAT session entry



NAT64 + DNS64 Limitations

- Only when the DNS is involved
 - I.e., hostnames are used
 - E.g., it does not work in case the user directly specifies an IPv4 address
 - E.g., `ping 1.2.3.4`
- No DNSSEC
 - In DNSSEC authoritative DNS signs record
 - But DNS64 modifies records

